

Weekly Progress Report

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Domain: Data Science and Machine Learning

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Week Ending: 04

I. Overview:

This week focused on developing the interactive Streamlit dashboard, integrating the trained machine learning model, implementing prediction modules, creating visualization components, and preparing the project for deployment and demonstration.

II. Achievements:

1. Streamlit Dashboard Development:

Project Name: Smart City Traffic Forecasting using Machine Learning

- Designed and developed a modern multi-page Streamlit dashboard.
 - Implemented responsive user interface with sidebar navigation.
 - Added professional dashboard layout with reusable UI components.
 - Developed Dark/Light mode support for better user experience.
 - Integrated project header, footer, and navigation system.
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2. Prediction Module Development:

Successfully implemented traffic prediction functionalities.

- Developed Single Traffic Prediction module.
- Implemented Bulk Prediction using CSV file upload.
- Integrated trained Random Forest model with Streamlit dashboard.

- Generated prediction output in CSV format.
 - Added download option for prediction results.
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3. Data Visualization:

Implemented multiple visualization components including:

- Feature Importance Chart
 - Prediction Distribution
 - Correlation Heatmap
 - Residual Plot
 - Dataset Overview
 - Model Performance Metrics
 - Traffic Prediction Summary
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4. Project Integration and Deployment:

Successfully integrated all project modules.

- Connected preprocessing pipeline with prediction module.
 - Linked trained model with Streamlit application.
 - Organized project into modular architecture.
 - Added logging and exception handling.
 - Prepared GitHub repository with proper documentation.
 - Created README, LICENSE, and professional project structure.
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III. Challenges:

1. Dashboard Development:

- Designing a responsive and user-friendly dashboard.
- Managing navigation between multiple pages.

- Integrating visualization components efficiently.
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2. Project Deployment:

- Managing GitHub repository structure.
 - Handling large model files using .gitignore.
 - Resolving dependency and package compatibility issues.
 - Organizing project files for deployment.
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IV. Learning Resources:

- Streamlit Documentation
 - Plotly Documentation
 - Matplotlib Documentation
 - Scikit-Learn Documentation
 - Git & GitHub Documentation
 - Python Logging Documentation
 - Machine Learning Deployment Tutorials
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VI. Additional Comments:

“Successfully completed the Streamlit dashboard, integrated the machine learning model, implemented prediction and visualization modules, and prepared the project for deployment. The Smart City Traffic Forecasting system is fully functional and ready for demonstration, further optimization, and cloud deployment.”